Holly Pasquinelli
Lab 7

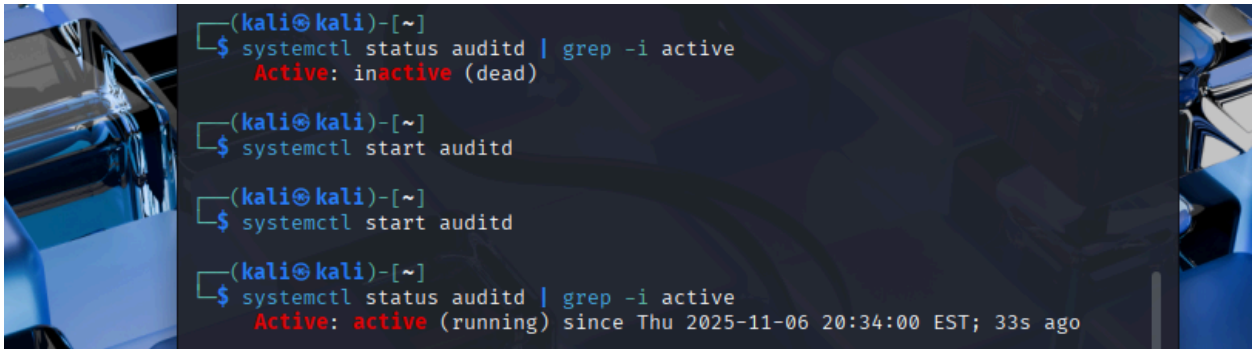
Linux Forensics (Linux Auditing System)

Non-technical Summary

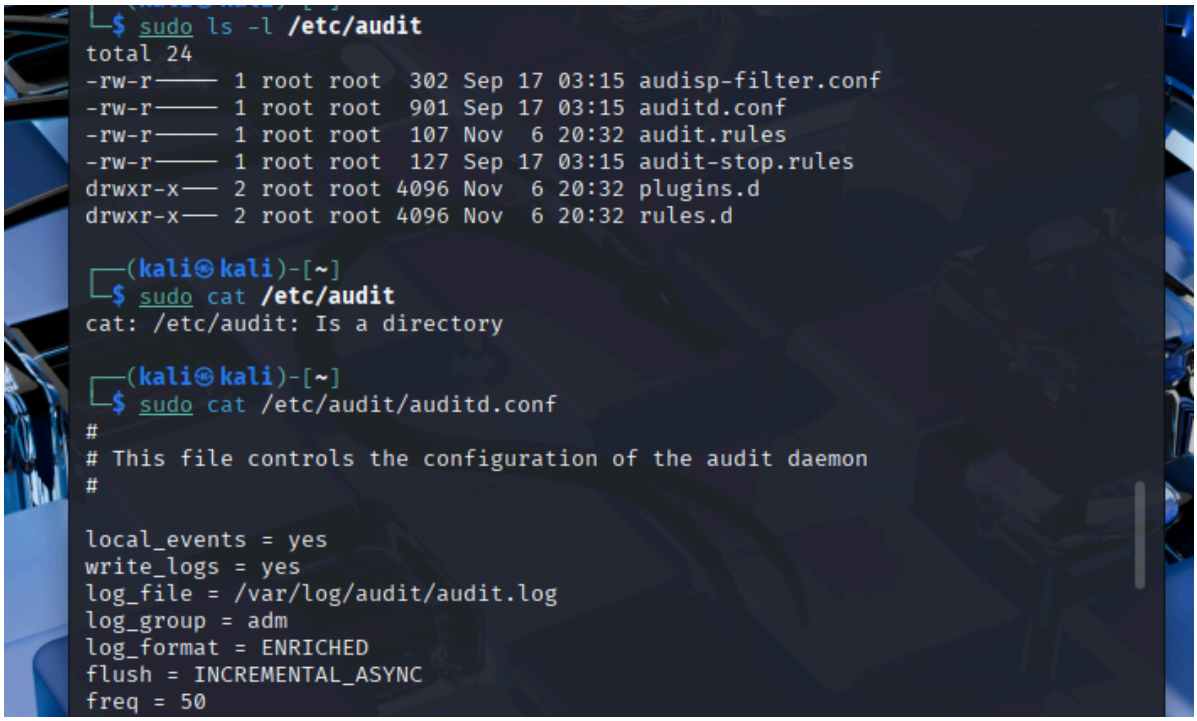
In this lab we explored the Linux auditing system using auditid and security auditing using Lynis. When using auditid, we are auditing logs being sent to the Kernel, authentication, and writing. This command is not related to security auditing, and we are not necessarily strengthening the system here. The audit system simply gives us an in depth detail of what is going on with the system. However, using Lynis is for security auditing of a computer. This is for system security, and for compliance testing. Lynis can also be used for penetration testing, vulnerability detection, and system hardening. Lynis gives an evaluation of security and the configuration of a system, and then gives warnings and recommendations on how to improve the system's security. This lab showed us how auditid generates the logs for Lynis to analyze and use for generating warnings and suggestions.

Technical Report

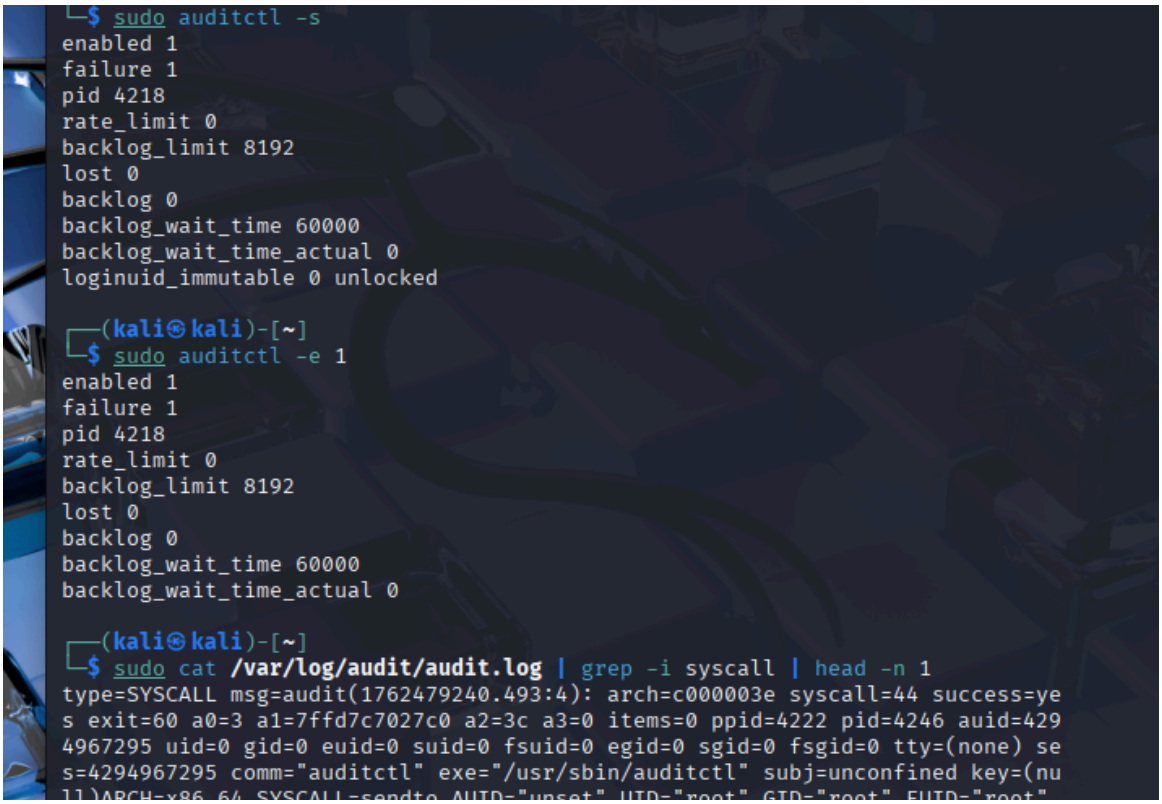
After we install auditid, we can check its status. We can see before we start it, it is inactive (dead). Once we start auditid, we can check the status again. We see that auditid is now active (running), and it gives us the date and time since it's been up.



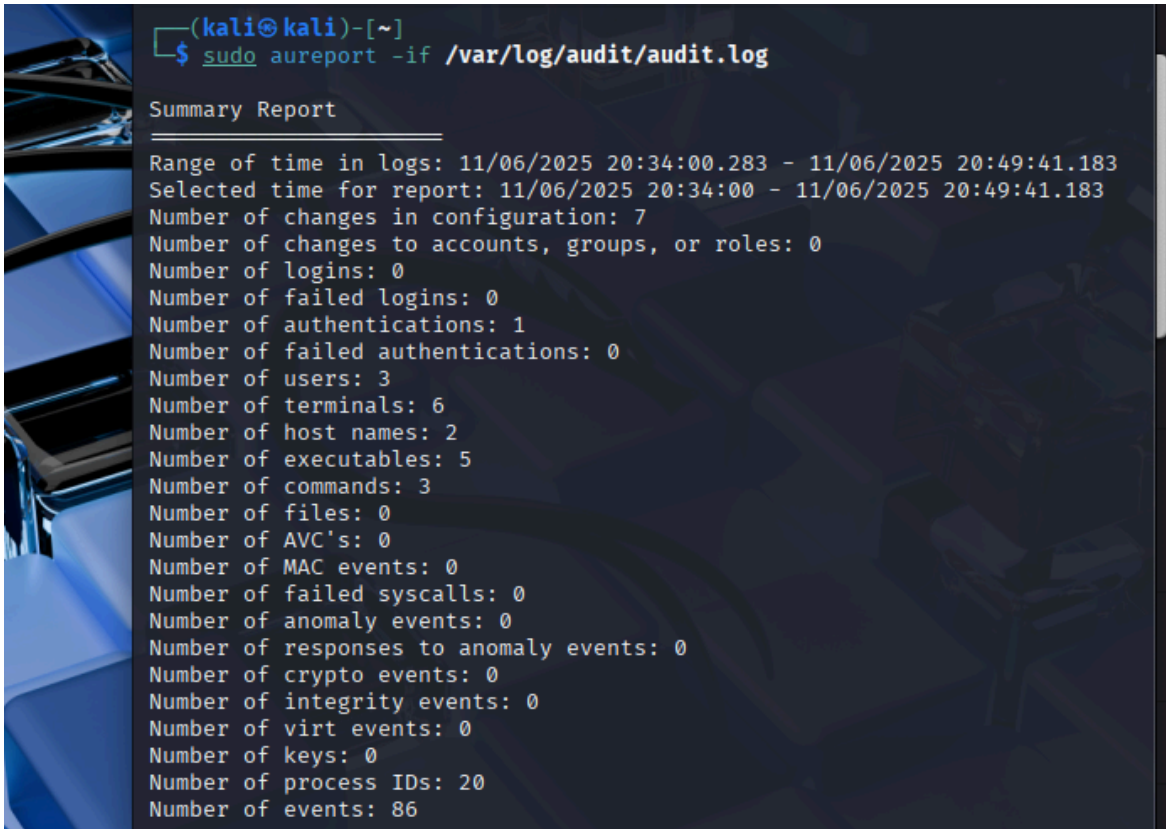
Next, we can use /etc/audit to access the auditid.conf file. This file controls the audit function that writes audit messages generated by the Kernel.



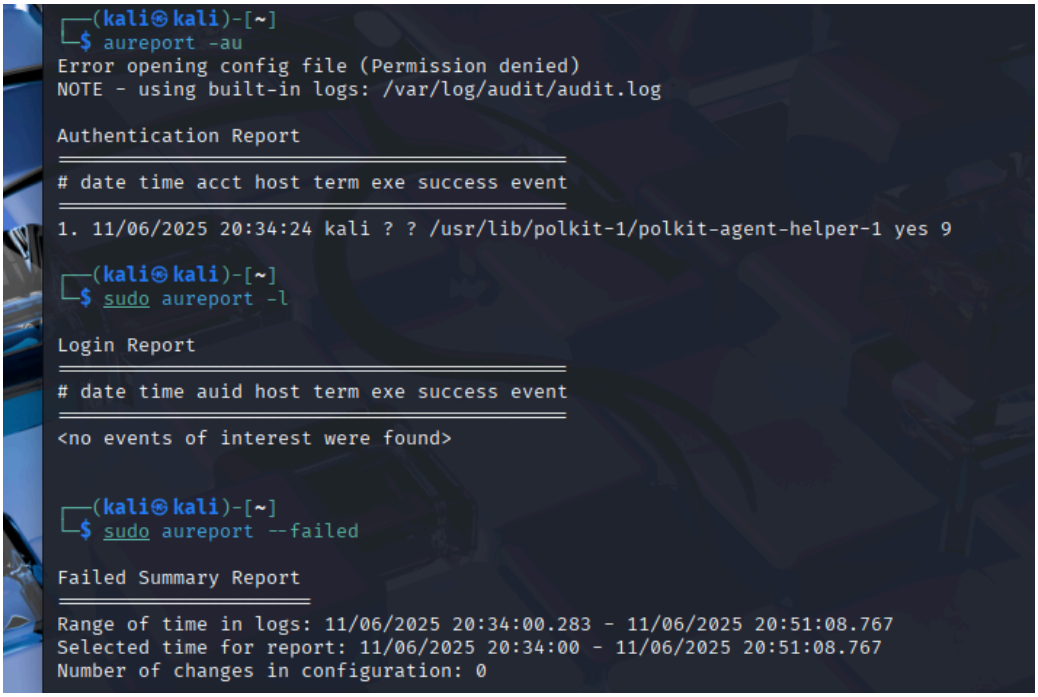
The auditctl tool controls and configures the audit interface. Below, we query the status of the audit daemon. We see a series of flags, such as if the system is enabled or disabled, failures, pid, limits, and backlogs. We enable the auditing system using sudo auditctl -e 1. We can then display one audit event by using sudo cat /var/log/audit/audit.log | grep -i syscall | head -n 1. We receive the type of event recorded, msg id, syscall trigger, cwd trigger, system boot trigger, and exe path.



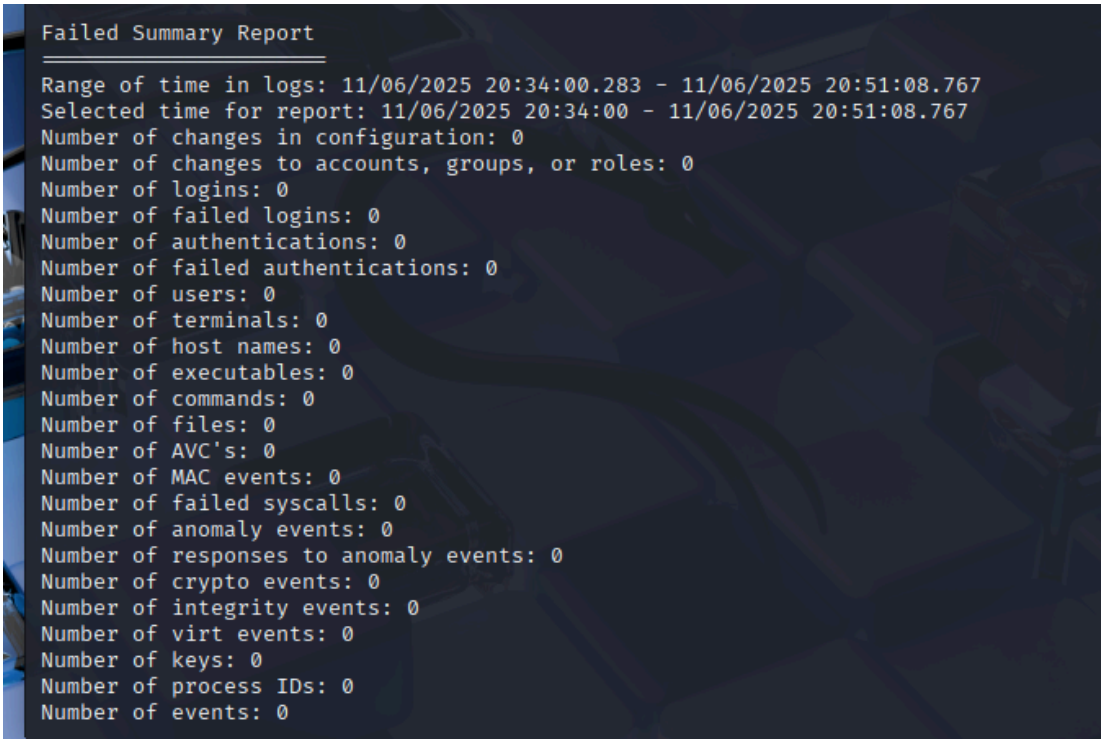
Using aureport, we created reports from the content /var/log/audit/audit.log. This is how we read audit logs from a file, and we can use -if to generate a detailed output.

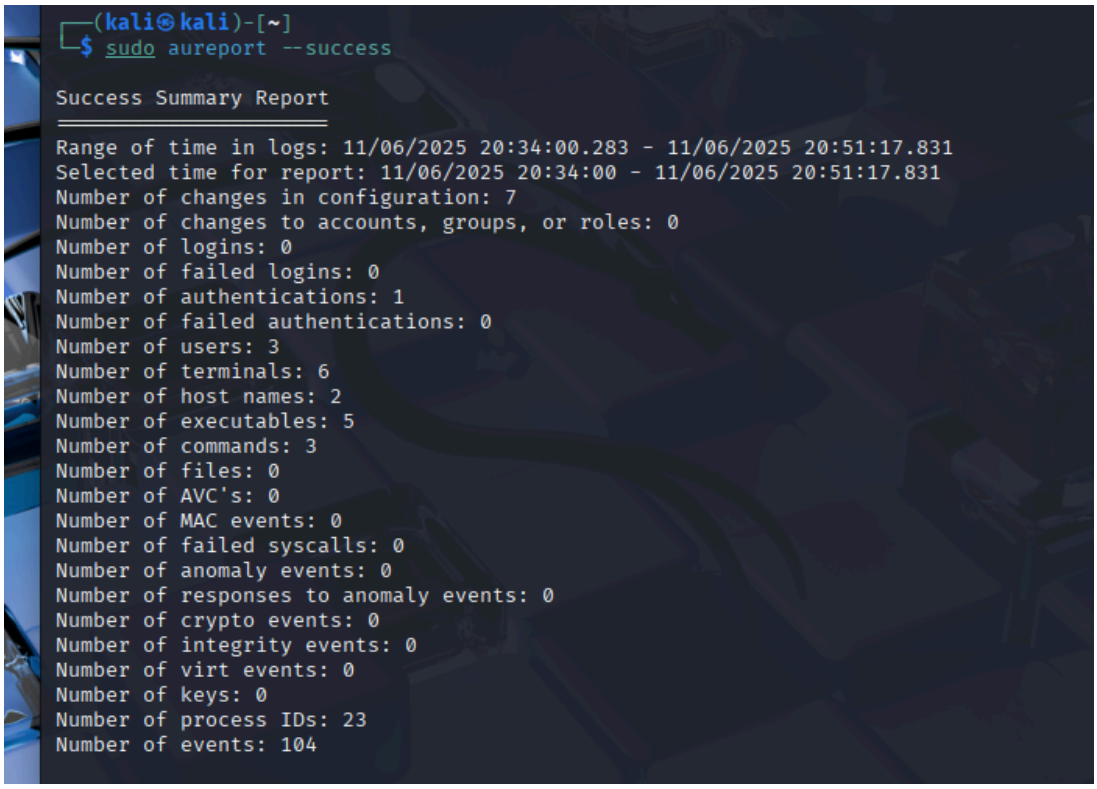


We can use -au to generate an authentication report showing us attempted authentication. We can also use -l to generate the login report, showing us attempted logins. Using --failed and --success we can display failed and successful events.

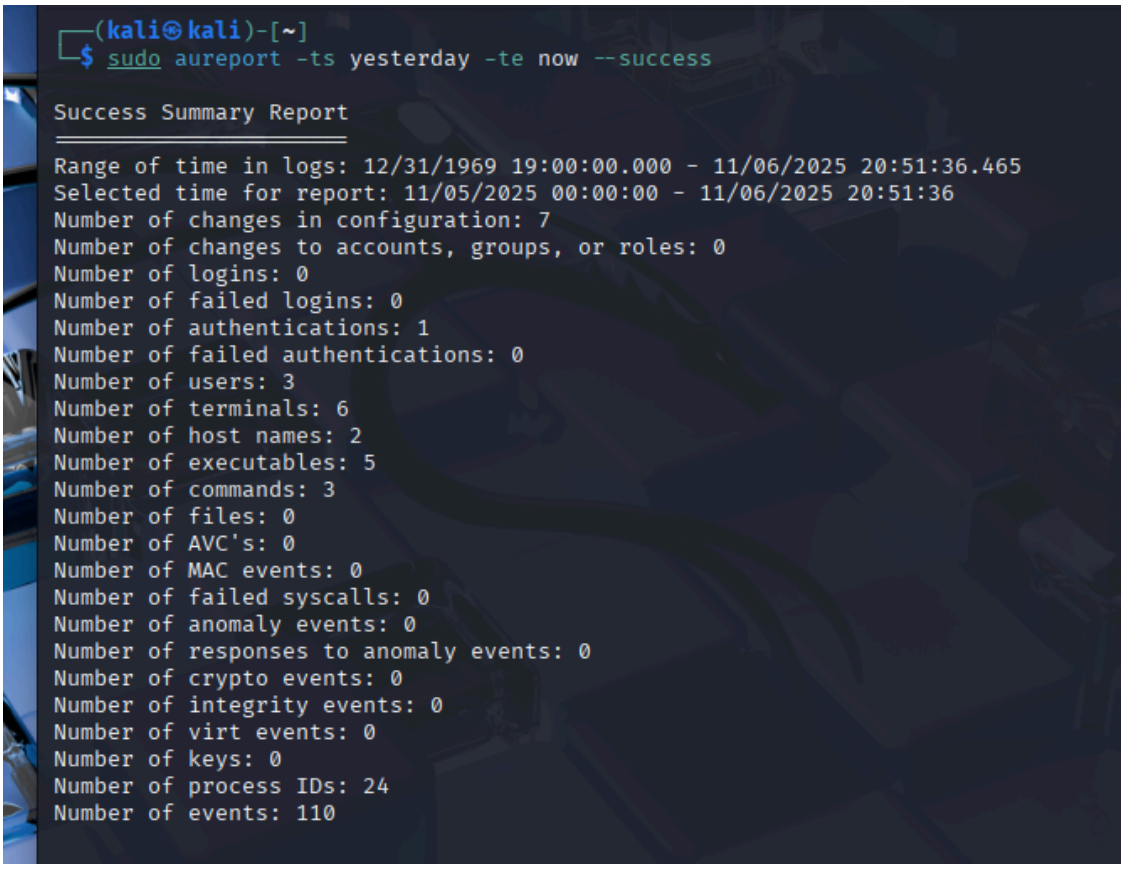


Below is the detailed failed events summary report generated by aureport, followed by the detailed successful events summary report.





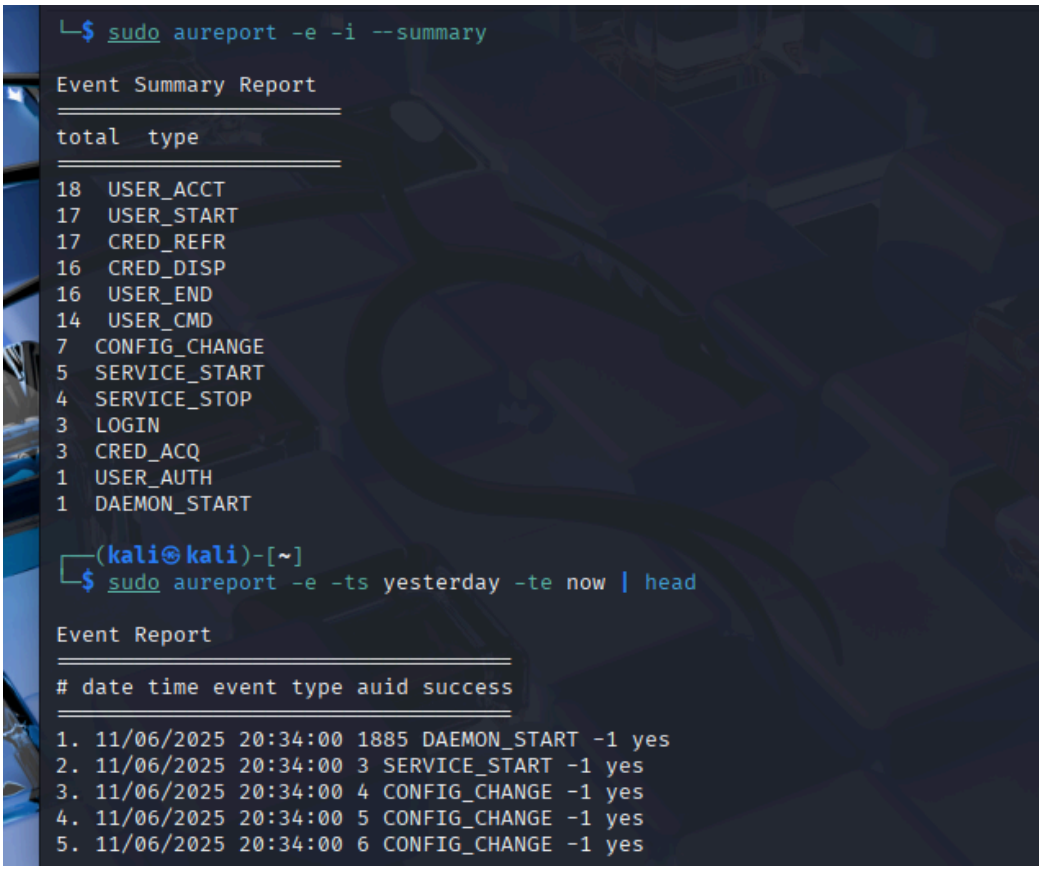
We can generate a more detailed report using -ts start time, and te end time.



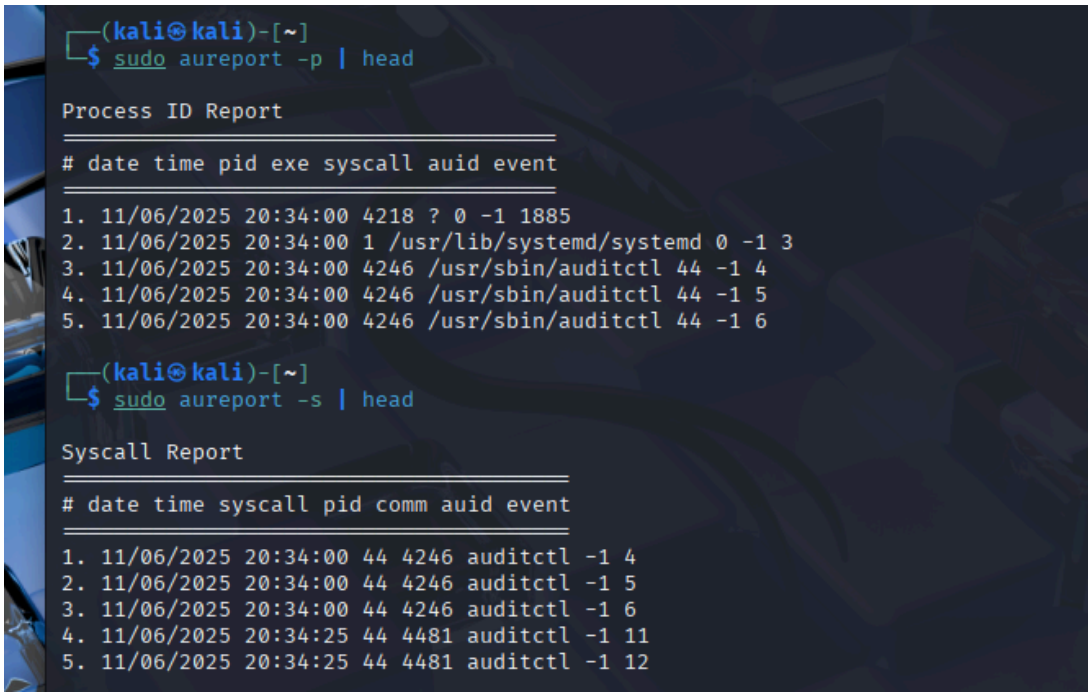
Using -u shows user reports and audit events. Using -l here will display usernames. For example, we see 1000 below, this would be displayed as Kali, and 0 would be displayed as root.



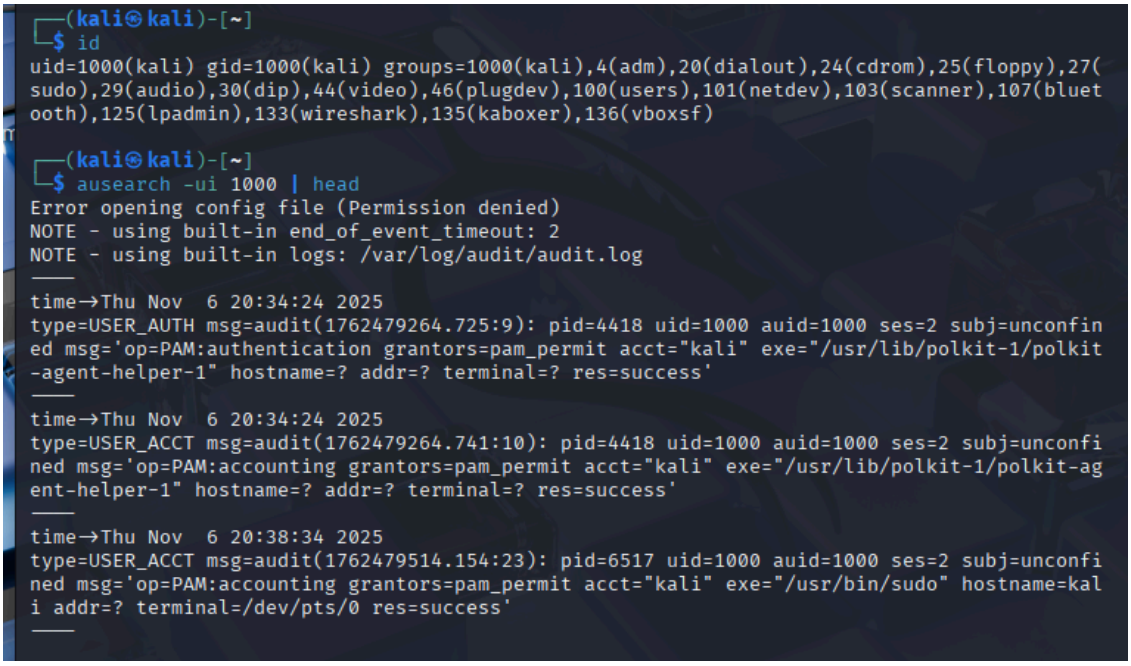
Below we generate the event summary report. We see totals for each type of event that has occurred. For example, CONFIG_CHANGE lets us know that someone has made changes to security configurations 7 times. Using additional formatting, we generate a more human readable report. This report shows us the number of the event, the date, time, type, auid, and if the event was successful, or if it failed.



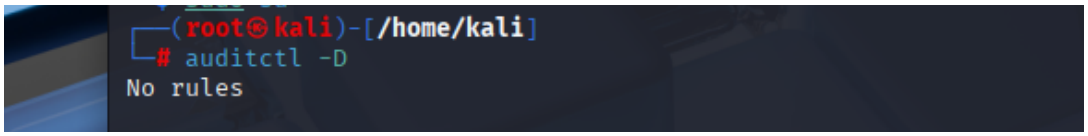
We can generate a similar report for process ID, and syscall reports.



Now, we can use ausearch to query the audit logs. id gets us the id of the current user that is logged in. Using ausearch we can search the events by a given user. Below, we see the user is Kali from the uid 1000.



Below, we try to delete the audit rules, but default there are no rules, and we see this below.



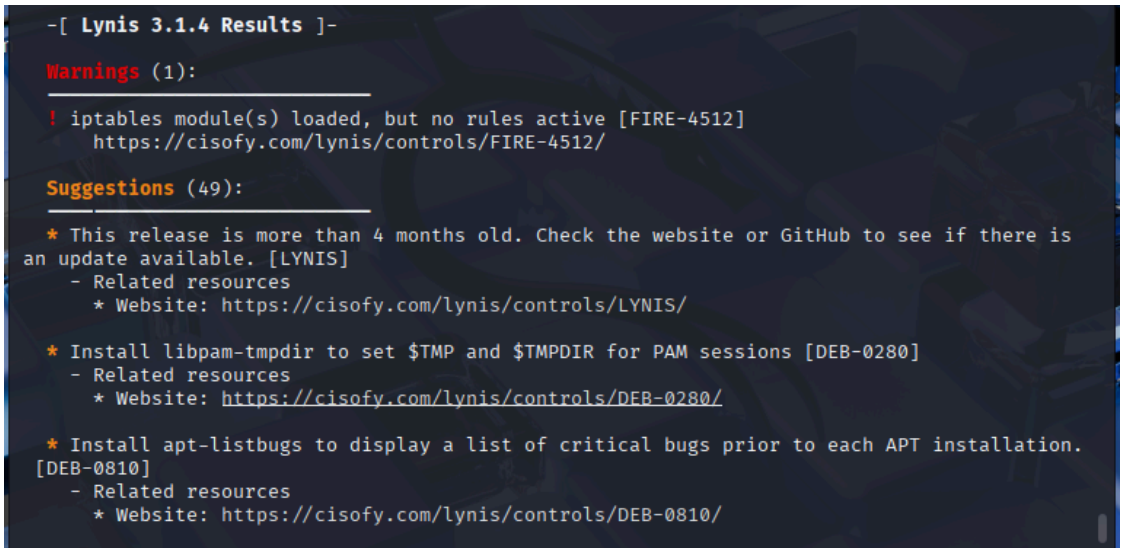
After Lynis is installed, we can display lynis commands.



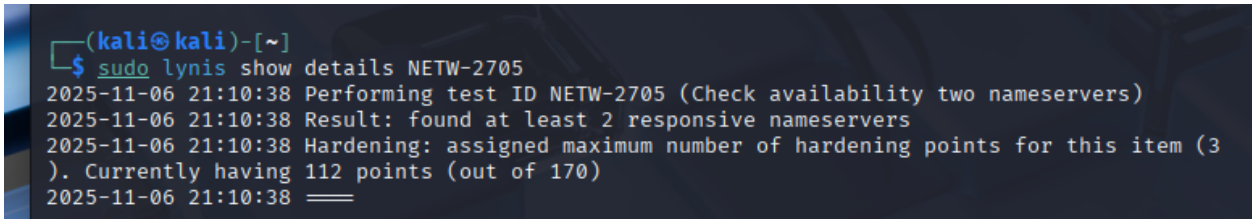
We can display Lynis settings using lynis show settings, using --brief hides descriptions.



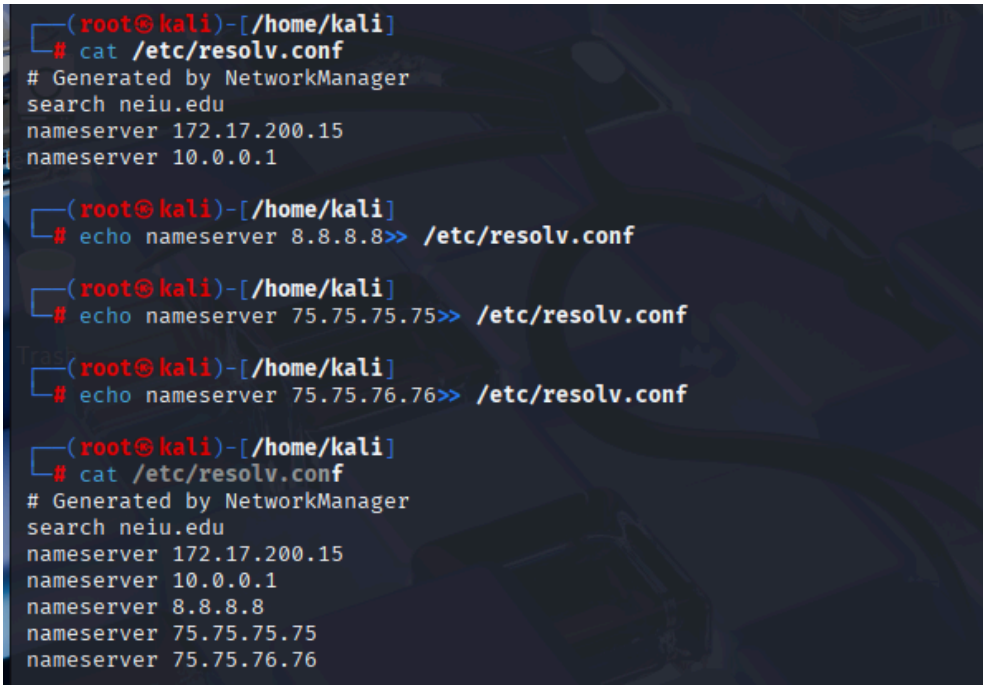
We can use sudo lynis audit system to complete the security audit of the system. Below, we see the report generated after the audit is complete. We see this audit reveals one warning, and 49 suggestions. The goal here is to try and resolve some of the suggestions to remove warnings.



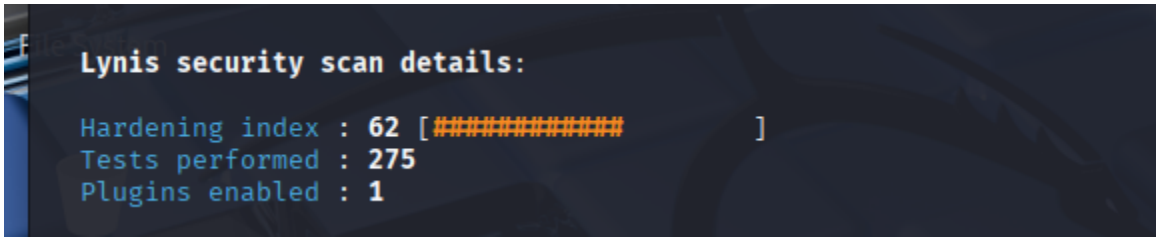
We can get more information about the audit by using the show details command. This also gives a hardening score.



We can display the content of the resolv.conf file and add more DNS servers. Once again displaying the content we see the added servers.



As mentioned above, we can try to resolve the suggestions to remove warings. This helps to improve the hardening index. We see this hardening score displayed below.



References

1. Lab 11 pdf